

Milestone 2: KPI Engineering & Dashboard Planning (Weeks 3–4)

Objective

The objective of Milestone 2 was to transform the cleaned military dataset into an analytical intelligence layer by engineering strategic KPIs and enriching it with metadata. Additionally, this milestone focused on designing the dashboard architecture, defining interaction logic, and preparing the foundation for full dashboard development.

This milestone bridged data engineering with visualization design.

Module 3: KPI Feature Engineering

Goal

To compute meaningful military performance indicators that enable comparative analysis across countries.

Dataset Refinement

Before KPI engineering, unnecessary columns related to energy production, reserves, debt, and merchant fleets were removed to focus analysis strictly on defense and military capability.

```
df.drop(columns=[col for col in columns_to_drop if col in df.columns],
inplace=True)
```

This reduced dimensional noise and improved dataset clarity.

Power Score (Composite Military Strength)

A composite strength score was calculated using:

- Active Personnel
- Tanks
- Total Military Aircraft
- Total Naval Fleet

```
df['Power_Score'] = (  
    df['active_personnel'] +  
    df['tanks'] +  
    df['total_military_aircraft'] +  
    df['total_naval_fleet']  
)
```

Purpose:

Creates a simplified representation of overall military strength.

Power Rank

Countries were ranked based on descending Power Score.

```
df["Power_Rank"] = df["Power_Score"].rank(ascending=False, method="min")
```

Purpose:

Enables global military ranking comparison.

Power Index Rank Gap

Measures the distance of each country from the top-ranked position.

```
df["Power_Index_Rank_Gap"] = df["Power_Rank"] - 1
```

Purpose:

Highlights strength gaps across nations.

Total Assets

Aggregated major combat assets:

```
df["Total_Assets"] = (  
    df["tanks"] +  
    df["total_military_aircraft"] +  
    df["total_naval_fleet"]  
)
```

Assets per Capita

Normalizes asset strength relative to population.

```
df["Assets_Per_Capita"] = df["Total_Assets"] / df["total_population"]
```

Purpose:

Indicates military intensity per citizen.

Budget-to-GDP Ratio

Measures defense spending relative to economic capacity.

```
df["Budget_to_GDP_Ratio"] = (  
    df["defense_budget_usd"] /  
    df["purchasing_power_parity_usd"]  
)
```

Purpose:

Identifies countries allocating significant economic resources to defense.

Metadata Enrichment

Continent Mapping

Countries were mapped to continents using a predefined mapping dictionary:

```
df["Continent"] = df["country"].map(continent_map).fillna("Unknown")
```

Region Classification

Region was aligned with continent for simplified segmentation:

```
df["Region"] = df["Continent"]
```

NATO Alliance Classification

Countries were categorized based on NATO membership:

```
df["Alliance"] = df["country"].apply(  
    lambda x: "NATO" if x in nato_countries else "Non-NATO"  
)
```

Final Export

The enriched dataset was exported:

```
df.to_csv("military_final.csv", index=False)
```

Module 3 Outcome

6+ KPIs successfully engineered
Composite Power Score and ranking created
Military asset intensity calculated
Economic allocation ratio computed
Geopolitical metadata added
Dashboard-ready dataset prepared

Module 4: Dashboard Planning & Prototyping

Goal

To design the structure, interaction flow, and analytical layout of the dashboard ecosystem before full implementation.

Dashboard Architecture Design

Four interconnected dashboard modules were planned:

1. **Quick Stats**
 - Global overview
 - Top 10 countries by Power Rank
 - KPI summary cards
 - Region & Alliance filters
2. **Nation Overview**
 - Country-level detailed analysis
 - Bar charts for manpower, aircraft, navy
 - Defense budget comparison
 - Interactive map view
3. **Compare Powers**
 - Side-by-side comparison of two countries
 - Metrics: manpower, aircraft, navy, budget
 - Ranking comparison

4. **Coalition Builder**

- Multi-country selection
- Aggregated coalition strength
- Comparison against a reference country